

CSA 1507 - Cloud Computing for big data analysis

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1. Define cloud computing and explain four characteristics

Definition: cloud computing is a model for delivering computing resources - services, with the provider owning and managing the underlying infrastructure.

Four distinguishing characteristics:

1. On demand and self service: use a provision computing resources automatically as needed, without requiring human interaction with the provider. on-premise computing, by contrast, requires manual procurement, installation.

2. Broad network access - cloud resources are accessible over the internet from a wide range of devices (laptop, phones, tablets).

3. Resource pooling and multi-tenancy: cloud provides pool computing resources to serve multiple customers using a multi-tenant model, dynamically is dedicated to a single organization.

4. Measured service: usage is metered, and customers pay only for what they consume. on-premise computing instead involves large upfront capital and expenditure regardless of actual utilization.

2. Trace the Evolution of cloud computing:

Hardware Evolution:

- * Main frames (1950-60s) centralized, expensive computers shared by many users via time-sharing an early precursor to the idea of shared computing resources.

- * Mini computers (1970s-80s)

- * Client Server computing (1980s-90s): Networked architecture where client machines requested services from centralized servers.

- * Commodity hardware and clusters (1990s-2000s)

The physical foundation that made large-scale data centres economically feasible.

Internet and software solution:

- * Growth of Internet (1990s): provided the ubiquitous network connectivity necessary to access remote resources easily:

- * Distributed computing and grid computing (1990s-2000s)

- * Virtualization software (2000s): Enabled multiple isolated virtual machines to run on shared physical hardware, decoupling software from hardware and multi-tenancy practical.

3. The Role of server virtualization in cloud computing.

Role of virtualization: server virtualization uses a hypervisor to partition a single physical server into multiple isolated virtual machines (VMS) each with its own operating system and resources and supporting the isolation needed for multi-tenancy.

virtualization vs cloud computing.

virtualization is an enabling technology: A technical mechanism for abstracting physical hardware into multiple logical / virtual resources. It can exist entirely within a single organization data center with no "cloud" involved at all.

Cloud computing is a service delivery model: It describes how computing resources are offered to users.

virtualization is one of the key technologies that makes cloud computing practical and cost effective, but cloud computing also depends on other elements that beyond virtualization alone.

4. Discuss the role of data centers in delivering:

Data centers are the physical backbone of cloud-computing - large facilities housing the servers, storage systems, and networking equipment that actually run cloud workloads.

=> physical infrastructure hosting: Data centers the thousand of physical servers, storage arrays, and networking hardware that cloud providers virtualize and offer as service.

=> Reliability and redundancy: cloud providers operate multiple geographically distributed data centers and so that if one facility fails, services can fall over to another, supporting the high availability cloud customers expect.

=> scalability: Because data centers massive pooled resources, they allow providers to rapidly allocate additional compute/storage capacity to customers on demand.

Security and compliance: compliance certifications that individual on premise setups often cannot match at the same cost.

Network connectivity: Data centers are built with high band-width, low latency connections to the internet backbone.

* Ensuring that cloud services delivered to users worldwide remain fast and responsive.

5. Differentiate IaaS, PaaS, SaaS and XaaS with one suitable real world ex: for each model.

Model	What the provider manages	What the user manages	Real world example:
IaaS - Infrastructure as a service	physical hardware, networking storage	OS, runtime, application, data	Amazon EC2, virtual machines
PaaS - Platform as a service.	Infrastructure + OS + runtime / middle ware	Application and the data	Google app, engine
SaaS - Software as a service	Everything - infrastructure and the application itself.	Just their data and user level	Google / workspace / sales force.
XaaS - Everything as a service	An umbrella term covering any IT capability a service	depends on the specific delivered	Database as a RDS or desktop as a service